Rudra Barot

Data Scientist

Harrison, NJ | 551-344-6924 | rudra.b@worksemails.com

SUMMARY

Data Scientist with 4+ years of experience in leveraging advanced analytics, machine learning, and AI to solve complex business challenges and deliver measurable value. Expertise in developing predictive models, natural language processing (NLP), and statistical analysis across diverse sectors including healthcare, finance, and technology. Proven track record of driving business improvements by harnessing data to improve decision-making, optimize processes, and reduce costs. Proficient in Python, R, SQL, and cloud platforms like AWS and Azure, with hands-on experience in deploying scalable solutions using Spark, Databricks, and BigQuery. Skilled in creating data-driven insights and visualizations through Tableau, Power BI, and ggplot2, with a focus on delivering actionable outcomes for stakeholders. Strong collaborator, thriving in Agile environments, and dedicated to enhancing data-driven strategies that lead to innovation and continuous growth.

SKILLS

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, R, SQL, SAS
IDEs & Development Tools:	Visual Studio Code, PyCharm, Jupyter Notebook
Cloud Technologies:	Google Cloud Platform, Azure, AWS
Bigdata Ecosystem:	Alteryx, Hadoop, Apache Spark, Kafka, Hive
Machine Learning Algorithms:	Linear/Logistic Regression, Decision Trees, Supervised Learning, Unsupervised Learning, Classification, SVM, Random Forests, Naive Bayes, KNN, K Means, CNN, RNN, and LSTM
Data Packages:	NumPy, Pandas, Matplotlib, SciPy, Seaborn, ggplot2, PySpark, Scikit-learn, TensorFlow, Keras, NLTK, OpenCV
Visualization Tools:	Tableau, Power BI, Excel, Qlik, Google Analytics, Meta Bar
Databases & Warehouses:	MySQL, PostgreSQL, SQL Server, MongoDB, Oracle, Snowflake
Financial Skills:	Statistical Modeling, Financial Analytics, Time Series Analysis, Portfolio Risk Management, Credit Risk Modeling
Other Skills:	Structured Data, Unstructured Data, Machine Learning, NLP, EDA, ETL Process, ELT Process, MLOps, Communication Skill, Meta Base, Databricks, Data Visualization, LLM Model, Predictive Analytics, Pattern Recognition, JMP, Data Integrity, Quantitative Data, Data Science, Statics, statistical analysis, Data Analytics
Operating System:	Windows, Linux

EXPERIENCE

Genworth, NJ, USA | Data Scientist — June 2023 - Current

- Operated in Agile environments to deliver iterative solutions, quickly adapting to evolving project requirements.
- Conducted hypothesis testing to evaluate variable relationships and inform data-driven decisions.
- Analysed structured and unstructured datasets, uncovering patterns and visualizing insights through interactive charts using ggplot2.
- Applied Naïve Bayes, Logistic Regression, Random Forest, and XGBoost models to accurately predict loan defaults.
- Implemented Support Vector Machine (SVM) algorithms for classification and regression tasks, identifying optimal hyperplanes for data classification and continuous outcome prediction.
- Designed and deployed recommender systems using collaborative filtering, scaled on AWS EMR, to suggest personalized courses.
- Leveraged Natural Language Processing (NLP) techniques to enhance customer satisfaction and optimize support processes.
- Conducted data analysis using Tableau to monitor application performance, improving response efficiency by 23%.
- Designed and optimized MySQL database schemas for efficient data storage and retrieval, enhancing application performance and scalability.

Druva Software, MH, India | Data Scientist — Jan 2020 - July 2022

- Developed, tested, and validated robust data science models and solutions using rigorous Waterfall methodologies.
- Leveraged Python libraries like SciPy, Scikit-learn, and Seaborn for advanced data analysis, uncovering patterns in user and product behavior.
- Developed and deployed machine learning models, achieving a 15% average accuracy improvement and reducing forecasting errors by 25%.
- Integrated Power BI with Excel and other data sources to create dynamic, auto-updating reports for real-time insights.
- Optimized healthcare supply chain logistics by analyzing demand patterns for pharmaceuticals and medical supplies, reducing inventory costs by 10% while ensuring timely availability.
- Worked with healthcare providers to optimize clinical workflows by analyzing patient flow data, resulting in a 12% improvement in appointment scheduling efficiency and patient throughput.
- Collaborated with cross-functional teams to integrate Databricks solutions into existing data pipelines, enhancing data flow and processing.
- Applied cutting-edge statistical, machine learning, and artificial intelligence techniques, including wave equation modeling and Latent Dirichlet Allocation (LLM), to address complex business challenges.
- Utilized AWS EC2 for scalable data processing, model training, and inference, optimizing performance for variable workloads.
- Built deep learning models (Keras, RNN, LSTM) for industrial decision support systems, optimizing operational efficiency.
- Ensured data integrity and availability by implementing robust backup and recovery strategies for SQL Server databases.

EDUCATION

Master in Computer Science | New Jersey Institute of Technology, Newark, New Jersey, USA

Bachelor in Information Science and Engineering | Jain University, Bangalore, Karnataka, India